DNA / Protein Sequence Analysis

Comparative BLAST Study: p53 (TP53) & KRAS Wild-Type

AUTHOR	DATE
Charles Arnaud Jesutin HOUNSOU-KAN	**May - June 20, 2026**
Medical Biotechnologist	

1. Introduction

The oncogenic axis formed by tumor suppressor **p53 (TP53)** and proto-oncogene **KRAS** constitutes a central pivot of mammalian tumorigenesis. p53 is inactivated in approximately 50% of all human cancers, while KRAS is the most frequently mutated oncogene, with a prevalence reaching 90% in pancreatic ductal adenocarcinoma (PDAC). Among KRAS alterations, the **G12D mutation** is dominant, representing 42% of PDAC cases [1,2].

1.1 p53 (TP53): Molecular Architecture and the "Guardian of the Genome"

Human p53 is a 53-kDa nuclear phosphoprotein of 393 amino acids, encoded by a 20-kb gene located on the short arm of chromosome 17, at cytogenetic position **17p13.1**, and organized into 11 exons [9]. Functionally, p53 sits at the centre of the cellular DNA-damage response, enforcing two principal cellcycle checkpoints the *G1/S transition*, through transcriptional induction of the CDK inhibitor $p21^{WAF1/CIP1}$, and the *G2/M transition* thereby preventing the propagation of damaged genomes into daughter cells [9].

When damage is irreparable or cellular stress is sustained, p53 shifts the **Bax/Bcl-2** rheostat in favour of the pro-apoptotic Bax protein, committing the cell to the *mitochondrial (intrinsic) apoptotic pathway* [9]. This dual capacity arresting the cycle to allow repair, or triggering elimination of irreparably damaged cells underlies its designation as the "guardian of the genome," a role that extends to safeguarding chromosomal stability by limiting the accrual of aneuploid cells [10]. p53 further behaves as a broadspectrum sensor of cellular stress, integrating signals as diverse as DNA damage, hypoxia, oncogene activation, nutrient deprivation, and ribosomal dysfunction [10].

Consistent with this gatekeeping function, TP53 is mutated or functionally inactivated in more than half of all human cancers [10], predominantly (≈85%) through missense substitutions clustered within the highly conserved central DNA-binding domain (exons 5–9), at recurrent mutational hotspots including R175, G245, R248, R249, R273 and R282 [9,11]. Because this core domain tolerates a wide diversity of substitutions while remaining under strong purifying selection at the amino-acid level across mammals, p53 provides an instructive evolutionary counterpoint to the near-absolute conservation of the KRAS catalytic core discussed below.

1.2 KRAS: A Molecular Switch Under Extreme Structural Constraint

KRAS belongs to the RAS superfamily of small GTPases, a class of molecular switches that cycle between an inactive, GDP-bound conformation and an active, GTP-bound conformation; only the GTPloaded state can engage downstream effectors such as RAF, PI3K and RalGDS [12]. The KRAS catalytic (G) domain is organized into two lobes: an **effector lobe** (residues 1–86), which contains the switch I and switch II regions together with the **P-loop** that coordinates the phosphate groups of the bound nucleotide, and an **allosteric lobe** (residues 87–171) implicated in membrane engagement [13]. The three catalytic residues most frequently altered in cancer glycine 12, glycine 13 and glutamine 61 all reside within this effector lobe [13].

Following translation, KRAS undergoes post-translational maturation of its C-terminal CAAX motif farnesylation, RCE1-mediated proteolytic cleavage and ICMT-dependent carboxymethylation which anchors the protein to the inner leaflet of the plasma membrane [14]. Once GTP-loaded, KRAS relays proliferative and survival signals principally through the *RAF–MEK–ERK (MAPK)* and *PI3K–AKT–mTOR* cascades [14]. Among the four RAS isoforms (HRAS, NRAS, KRAS4A, KRAS4B), KRAS is by far the most frequently mutated in human malignancy, accounting for the large majority of oncogenic RAS alterations observed across cancer types [12]. Together with the extreme structural constraints imposed on its P-loop detailed in Section 4.2 this establishes the biological rationale for the comparative conservation analysis presented below, in particular for the GTPase P-loop position G12.

Table 1. Proteins analyzed in this study.

Protein	Gene	UniProt ID	Length	Biological Role
p53	TP53	P04637	393 aa	Tumor suppressor - Guardian of the genome
KRAS (WT)	KRAS	P01116	189 aa	Proto-oncogene GTPase / cell proliferation

Building on this molecular framework, the present study uses **BLAST (Basic Local Alignment Search Tool)** to quantify the evolutionary conservation of both proteins across mammalian species, validating the extreme structural constraints acting on their functional domains particularly the GTPase P-loop of KRAS containing position G12.

2. Materials and Methods

2.1 Sequence Retrieval

Canonical protein sequences were retrieved programmatically using the UniProt REST API (rest.uniprot.org) in FASTA format. Biopython 1.85 SeqIO module was used to parse and validate the records [3].

- p53: https://rest.uniprot.org/uniprotkb/P04637.fasta
- KRAS: https://rest.uniprot.org/uniprotkb/P01116.fasta

2.2 BLAST Analysis

Protein BLAST (blastp) queries were submitted to NCBI servers via Biopython's Bio.Blast.NCBIWWW.qblast() function with the parameters below.

Table 2. BLAST search parameters.

Parameter	Value
Program	blastp (protein-protein BLAST)
Database	refseq_protein
Filter	Mammalia [Organism] (taxid: 40674)
Max hits	15 per query
Tool	Biopython 1.85 / NCBIWWW.qblast()

3. Results

3.1 p53 (TP53) — BLAST Results

The BLAST search for human p53 (393 aa) retrieved 15 significant alignments, all with E-values of 0.0, confirming extremely high statistical significance. The mean identity across top hits was 98.0% (Min: 95.7% | Max: 100%). The closest homologs were found in great apes (Pan paniscus at 100%, Gorilla gorilla at 99.75%), consistent with known phylogenetic relationships.

Table 3. Top 5 BLAST hits for p53 (TP53) against Mammalia.

Organism	% Identity	% Similarity	Score (bits)	E-value
Pan paniscus	100.0%	100.0%	2101	0.0
Gorilla gorilla gorilla	99.75%	99.75%	2103	0.0
Pongo abelii	97.71%	98.22%	2059	0.0
Nomascus leucogenys	98.0%	98.5%	2058	0.0
Macaca thibetana thibetana	95.7%	97.2%	2026	0.0

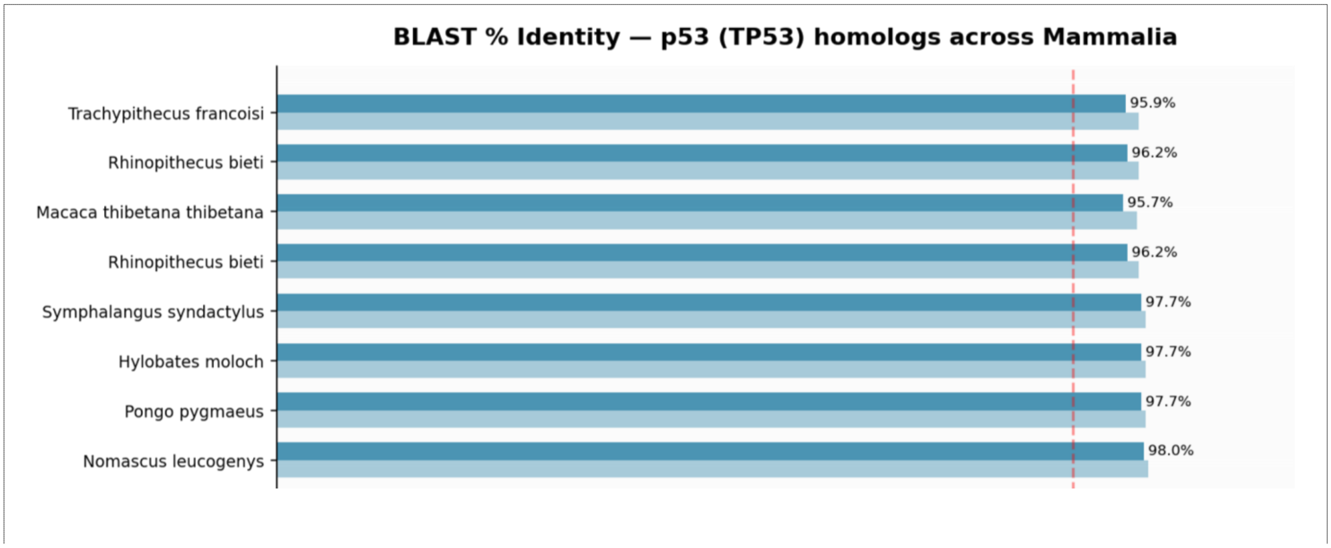

Figure 1: BLAST Identity p53 (TP53) homologs across Mammalia

3.2 KRAS Wild-Type – BLAST Results

The BLAST search for human KRAS (189 aa) retrieved 15 alignments with even higher conservation than p53. Remarkably, 6 out of 15 top hits showed 100% sequence identity across phylogenetically distant species including bats, dolphins, and primates. Mean identity: 99.7% (Min: 99.5% | Max: 100%).

Table 4. Top 5 BLAST hits for KRAS wild-type against Mammalia

Organism	% Identity	% Similarity	Score (bits)	E-value
Myotis nattereri	100.0%	100.0%	1003	1.43e-138
Gorilla gorilla gorilla	100.0%	100.0%	1006	2.06e-138
Macaca fascicularis	100.0%	100.0%	1006	2.06e-138
Desmodus rotundus	99.5%	100.0%	1005	2.67e-138
Giraffa tippelskirchi	99.5%	100.0%	1005	3.56e-138

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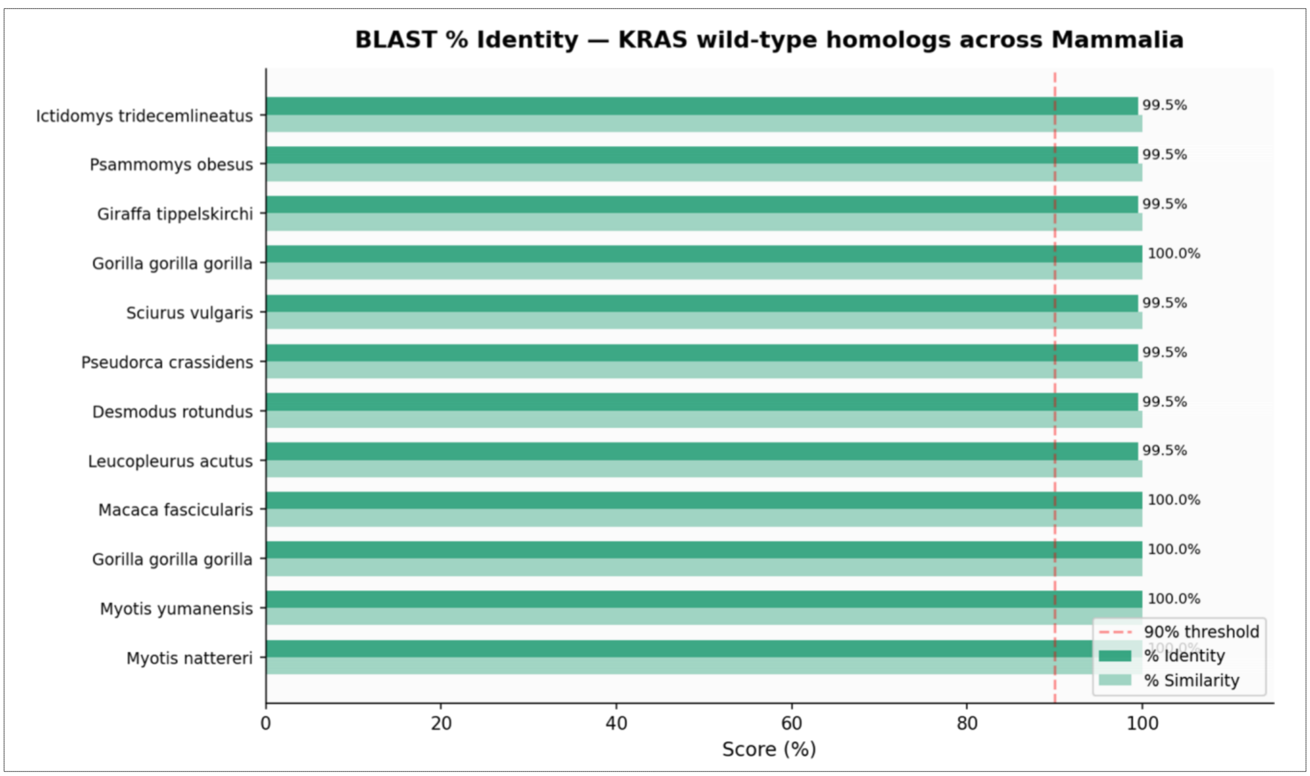

Figure 2: BLAST Identity KRAS wild-type homologs across Mammalia

3.3 Comparative Conservation Summary

Table 5. Comparative conservation summary: p53 vs. KRAS wild-type.

Metric	p53 (TP53)	KRAS (WT)	Interpretation
Mean % Identity	98.0%	99.7%	KRAS more conserved than p53
Min % Identity	95.7%	99.5%	Both extremely high minimums
Max % Identity	100.0%	100.0%	Perfect identity in closest species
Hits at 100% ID	2/15	6/15	KRAS more widely identical
All E-values	0.0	~10e-138	Both highly statistically significant
Protein length	393 aa	189 aa	p53 ~2x longer

4. Discussion

4.1 Comparative Evolutionary Constraint

The BLAST analysis confirms that both p53 and KRAS are highly conserved across Mammalia, reflecting their indispensable roles in fundamental cellular processes. KRAS proves significantly more conserved than p53 (99.7% vs. 98.0% mean identity), a finding that underscores the extreme functional constraints acting on its GTPase domain, relative to the comparatively greater structural and functional plasticity tolerated by the p53 sequence outside its core DNA-binding domain.

4.2 Structural Basis of G12 Ultra-Conservation and the G12D Oncogenic Switch

The ultra-conservation of glycine at **position 12 (G12)** within the P-loop of the KRAS GTPase domain constitutes the central biological insight of this analysis. The P-loop is a glycine-rich phosphate-binding motif that cradles the β- and γ-phosphates of the bound guanine nucleotide, and G12 sits at its apex, immediately adjacent to the catalytic machinery governing GTP hydrolysis [13]. Because glycine lacks a side chain, it introduces no steric hindrance within this tightly packed pocket, leaving just enough conformational space for the water molecule that bridges the γ-phosphate during both intrinsic and GAPstimulated hydrolysis [13].

The pathological **G12D substitution** (glycine → aspartic acid) introduces a bulkier, negatively charged side chain that structurally displaces this catalytic bridging water molecule [13]. This single steric and electrostatic perturbation is sufficient to cripple both the intrinsic GTPase activity of KRAS and its ability to dock productively with *GTPase-Activating Proteins (GAPs)* such as neurofibromin (NF1): mutants at G12 like those at Q61 are rendered insensitive to GAP-mediated catalysis [14]. Deprived of GAP assistance, mutant KRAS cannot complete hydrolysis of GTP to GDP and becomes locked in its constitutively active, GTP-bound conformation, driving unrestrained downstream MAPK and PI3K/AKT signaling and uncontrolled cell proliferation [2,13].

At the nucleotide level, this substitution most commonly arises from a G→A transition at the second base of codon 12, the single most frequent mutational event across the entire KRAS mutational spectrum, accounting for roughly 43% of all reported KRAS alterations [13]. This mechanistic fragility of the Ploop toward even a minimal side-chain change explains why purifying selection has preserved glycine at this position with near-absolute fidelity across the ~450 million years separating the mammalian species examined in this study (Table 4), and why a single point mutation is sufficient to convert a tightly regulated molecular switch into a constitutively oncogenic driver accounting for the dominance of G12D in ~42% of pancreatic ductal adenocarcinoma (PDAC) cases [1,2,5].

4.3 The p53 - KRAS Functional Axis: From Independent Conservation to Oncogenic Cooperation

Beyond their independent evolutionary trajectories, p53 and KRAS are mechanistically and clinically intertwined. In murine models of pancreatic carcinogenesis, oncogenic Kras-G12D alone is sufficient to initiate low-grade pancreatic intraepithelial neoplasia (PanIN), but progression to invasive, metastatic adenocarcinoma requires the additional loss or mutation of p53 as demonstrated in the widely used *KPC mouse model* ($Kras^{LSL\text{-}G12D/+}$; $Trp53^{R172H/+}$; Pdx1-Cre), in which roughly 80% of animals develop metastatic disease, compared with markedly reduced invasiveness in mice carrying only a conditional Trp53 knockout allele [2,15].

This genetic cooperativity is mirrored in human PDAC, where TP53 *missense* mutations rather than truncating mutations are associated with earlier invasiveness, higher rates of lymph-node involvement, and reduced disease-free and overall survival specifically within a KRAS-mutant genomic background [15]. Such observations reinforce the view that the ultra-conservation of the KRAS catalytic core documented here (Table 4) and the comparatively broader mutational tolerance of p53 (Table 3) are not merely parallel evolutionary phenomena, but reflect two complementary layers of a single oncogenic safeguard one structural (the GTPase switch) and one transcriptional (the genome guardian) whose simultaneous failure is a recurring signature of aggressive malignancy.

4.4 Therapeutic Implications

These results establish the biological foundation for bioinformatics-guided therapeutic strategies targeting the KRAS switch-II pocket. **MRTX1133**, the first selective non-covalent KRAS G12D inhibitor (IC^{50} < 2 nM), was identified through structure-based drug design exploiting this pocket [4]. **Zoldonrasib (RMC-9805)**, a covalent RAS(ON) inhibitor, demonstrated a 30% objective response rate and an 80% disease control rate in PDAC patients in Phase I data (2024) [6]. Additional G12D-directed agents are progressing through early clinical development, including HRS-4642 and the KRAS-G12Dtargeted protein degrader ASP3082, reflecting a rapidly maturing therapeutic pipeline built directly on the structural vulnerabilities of the P-loop described above [2].

5. Conclusion

This BLAST analysis demonstrates that the p53/KRAS axis has been subject to rigorous purifying selection for over 450 million years. KRAS is significantly more conserved than p53 (99.7% vs. 98.0%), with 6/15 top mammalian hits at perfect 100% identity a level of constraint directly traceable to the indispensable role of glycine 12 within the P-loop of its GTPase domain [13]. This evolutionary evidence validates the functional indispensability of the KRAS catalytic core and explains why a single point mutation (G12D), by displacing the catalytic water molecule required for GTP hydrolysis and abolishing GAP engagement, is sufficient to trigger oncogenic transformation a vulnerability preserved across every mammalian species examined and now actively exploited by structure-based therapeutics such as MRTX1133 and zoldonrasib [4,6,13].

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